Predicting AI Performance on Unseen Benchmarks: Shrinkage, Hierarchical Refinements and Their Limits

Shashwat Srivastava
Accenture
shashwat1322001@gmail.com
Team S2WISH
Code: https://github.com/Shashwat-srivastav/s2wish-paiec2026

Abstract

On benchmarks a model has never seen, shrinkage is the working core; what remains is item difficulty, which none of the label-free signals we tested without a runtime LLM supplies usefully (runtime LLM priors, which the rules permit, were not tested). We show this for the AIMS Predictive AI Evaluation Competition, which scores predictions on a hidden benchmark by the Brier area under the learning curve (ALC; lower is better). In leave-one-benchmark-out evaluation, empirical-Bayes shrinkage of early estimates gives 33.7 of the 36.3 thousandths that our small hierarchical Bayesian model gains over the official baseline. A pre-registered check on 89 unseen public benchmarks confirms the shrinkage gain; there the check detected no difference between the hierarchical model and shrinkage alone (inconclusive at the pre-registered $\pm 1 \times 10^{-3}$), so the refinements are unproven. On the platform draws we observed, no pair-level adjustment could improve our model by more than about 0.022 ALC (95% CI up to 0.033), in-sample. We contribute floor and ceiling bounds that locate the remaining loss in item difficulty; error bounds and exact replays for reading the platform's per-pair feedback; and negative results from 60 logged experiments (22 pre-registered).

1 Introduction

On unseen benchmarks, shrinkage does almost all the work; what remains is item difficulty, which none of the label-free signals we tested without a runtime LLM supplies usefully. Runtime LLM priors, which the rules permit, were not tested. Figure 1 shows the story in one picture.

Predicting outcomes is cheaper than running every AI system on every benchmark. The AIMS Predictive AI Evaluation Competition [1, 2] poses this under the hardest condition: the benchmark is new, so no system's responses to its items are known. Item response theory (IRT) methods need exactly those responses (§2). We therefore built a model that learns only what transfers across benchmarks.

What we show.

- **Shrinkage does almost all the work** (§5). The hierarchical refinements are not established across benchmarks; a pre-registered out-of-sample check detects no difference from shrinkage. Table 4 lists every rejected candidate.
- **The remaining loss is item difficulty** (§8). Oracles and floor and ceiling bounds show that pair-level calibration cannot close the gap; no label-free signal we tested without a runtime LLM usefully supplies item difficulty.
- **Per-pair feedback can be read exactly** (§6). We give error bounds, the reflection pitfall and exact replays (Appendix D); they reveal that platform pairs are harder than our local folds.

Preprint.

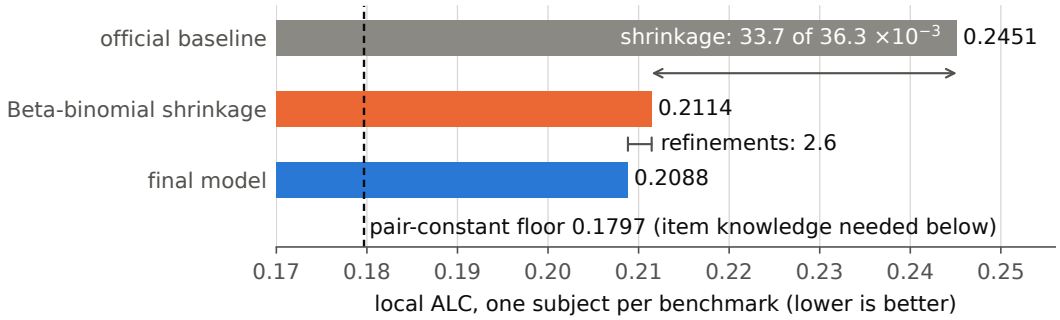


Figure 1: Local ALC with one subject per benchmark, mean over four held-out benchmarks. Beta-binomial shrinkage gives 33.7 of the 36.3×10^{-3} gain over the official baseline; the hierarchical refinements add the rest. Below the dashed pair-constant floor $a(1-a)$, the score of knowing each pair's accuracy but nothing about its items, only item-level knowledge helps (§8). Sources: `make_figures.py`; `experiments/runs/review_round1/ladder_intervals.json` and `floor_ceiling.json`.

2 Related work

IRT and efficient evaluation. tinyBenchmarks [3], metabench [4], Anchor Points [5], Fluid Benchmarking [8], amortized calibration [6] and IRT scaling laws [7] all calibrate items on many systems' responses to the target benchmark, which a hidden benchmark does not allow; contextual IRT with text embeddings ranks items at chance level across scenarios [9], as our text features do (Table 5). **Unseen data.** Instance-level assessors degrade out of distribution [10, 11]; forecasting benchmark accuracy from a task description needs an LLM at prediction time [12]; system-level abilities do transfer across benchmarks [13]. **Empirical Bayes.** Shrinkage of many small-sample rates toward a pooled level beats the raw mean [14, 15], and heavy-tailed weakly informative priors are a robust default for logistic models [16]. **Scoring and calibration.** The Brier score [17] is strictly proper [18]; Murphy's decomposition [19] separates reliability from resolution; temperature scaling [20] is the standard recalibration. **Acquisition and noise.** Active testing [21] and active inference [22] reduce to uniform sampling when items cannot be told apart, and adaptive testing on noisy item parameters can hurt [23]; small-sample evaluations need wider intervals than the central limit theorem suggests [24, 25], and repeated runs on the same item vary [26].

3 Task, evaluation and data

A *subject* is one AI-system configuration, described by eight strings: model name, provider, release and access dates, harness, harness version, reasoning effort and extras. An *item* is described by four: content, features, interactors and an anonymous benchmark ID. A submission implements `predict(input, labeled)` $\to p$ and may implement a label-acquisition hook. For each subject–benchmark pair the platform splits the items 50/50 into an acquisition pool and an evaluation pool. It reveals up to 31 acquisition labels and scores the evaluation pool at six label budgets:

$$\mathrm{ALC} = 0.1B_0 + 0.2B_1 + 0.2B_3 + 0.2B_7 + 0.2B_{15} + 0.1B_{31}, \tag{1}$$

where B_n is the Brier score, the mean of $(p-y)^2$, with n labels revealed. ALC is averaged over pairs with equal weight, so always predicting 0.5 scores 0.25.

At every budget, `labeled` holds the revealed answers of *every* sampled pair. Another subject's label on the same hidden benchmark, or on the same item, is therefore legal evidence.

We re-implemented the evaluator and drove the organisers' unmodified streaming client through it. After we fixed nine discrepancies in the first version, per-pair, per-budget Brier scores matched exactly (maximum difference 0). Each formative submission scores a fresh draw of about 1,000 items. In practice that is eight or nine pairs over seven anonymous benchmarks, usually with one subject per benchmark.

Table 1: Public training data (revision `bc8204d811`). MMDocRAG (not binary) is unused; SWE-rebench (one subject) is used for training only. Sources: `scripts/reproduce_all.py --only T1`, from the gated `data/*/response.parquet`.

benchmark	subjects	items	responses	pairs ≥ 80 items	mean pair accuracy
MathArena	104	1,755	92,944	81	0.61
Multi-SWE-bench	82	2,126	57,975	82	0.22
REAL web agents	33	233	4,449	26	0.34
ResearchCodeBench	31	212	6,784	31	0.35
SWE-rebench	1	6,306	67,074	1	0.48

Validation protocol and notation. Benchmarks are held out whole, so the only meaningful validation is leave-one-benchmark-out (LOBO). That gives four folds (Table 1: Multi-SWE-bench, REAL, ResearchCodeBench, MathArena), so every benchmark-level statement in this report rests on four numbers. Within a fold we draw test episodes in one of three ways:

- **one subject per benchmark**, which is what the platform's draws turned out to be. This is our primary setting, and every local number is in it unless stated otherwise;
- **platform-like draw**: about 1,000 items sampled as the platform does;
- **all subjects**: every subject present. This is an upper bound only, because it overstates cross-subject pooling about sevenfold.

All comparisons are paired on identical episodes. We quote differences in ALC in thousandths, written $\Delta\text{ALC} \times 10^{-3}$: a value of -1.0 means 0.001 lower (better) ALC. Unless stated otherwise, intervals are 95% bootstrap intervals over pairs. ρ_S denotes Spearman rank correlation. Research notes cited by number are in `docs/research_notes/` of the code release.

Where the predictable signal is. Per-benchmark Rasch fits place more logit-scale variance on items than on subjects for every benchmark. Re-running the same subject on the same item accounts for a further 20–27% of response variance (repeated cells in MathArena and SWE-rebench; note 11), which no predictor can remove. Subject metadata explains 16% of ability variance on an unseen benchmark (ridge R^2 under LOBO; note 14), mostly through release date. Model ranks transfer between coding benchmarks (ρ_S 0.6–0.9), but levels do not (intraclass correlation 0.22). Text features rank items within a benchmark but do not transfer to new benchmarks; an LLM difficulty rating, computed offline, transfers weakly (Table 5, Appendix B). A leakage audit found no runtime-visible field that predicts outcomes beyond the benchmark mean. It also showed that folds must be grouped by item content rather than item ID, because duplicates inflate MathArena by 0.014 Brier.

4 Method

The predictor is a Rasch-type model with four latent quantities per prediction:

$$\text{logit}\, P(y = 1) = \mu_b + \theta_{sb} - \beta_i + \gamma_{g(i)}, \tag{2}$$

$\mu_b \sim t_3(G/2,\ s)$	benchmark offset; evidence: every subject's labels on benchmark b
$\theta_{sb} \sim \mathcal{N}(0, \tau^2)$	pair ability; evidence: the subject's own labels
$\beta_i \sim \mathcal{N}(0, \sigma_\beta^2)$	item difficulty; evidence: other subjects' labels on the same item
$\gamma_g \sim \mathcal{N}(0, \lambda\sigma_\beta^2)$	item-group effect; evidence: the subject's labels in the same group.

Fitted on the five public benchmarks, $G = -0.58$ (the mean logit accuracy), $s = 0.78$, $\tau = 1.13$ and $\sigma_\beta = 2.70$; λ is the share of item-difficulty variance assigned to groups. Four design choices matter.

- **A hedged offset prior.** The offset prior is weakly informative and heavy-tailed [16], centred at $G/2 = -0.29$, halfway between the pooled level and 50%. This cuts worst-case regret on unusually easy benchmarks by 29% (platform-like draws) to 50% (one subject per benchmark) at no mean cost.
- **Numerical integration.** The posterior is integrated by deterministic numerical quadrature: a 121-point grid over μ and Gauss–Hermite quadrature over θ and β, with no sampling. In an early development run on SWE-rebench, plug-in point estimates reached 0.49 Brier after one label (research record, overview of design decisions).
- **Item pooling.** Items that share a key are pooled across subjects through the false-merge model and the optional key guard (Table 2).
- **Item groups.** Groups come from the fields of `item_features`, with λ averaged over three values by marginal likelihood.

Predictions are clipped to $[0.005, 0.995]$. Labels are acquired at random. Four non-random policies gained at most 0.27×10^{-3}, as theory predicts when the predictor cannot tell unlabelled items apart [21, 23]. The archive is pure Python and NumPy (three files, about 5 KB zipped) and takes 3.3–222 ms per call.

We built the model as a ladder of versions (Table 2). The *final model*, our selected submission, is the item-group model with the key guard switched off; its two code files are byte-identical to the item-group model's. Archive names and checksums are in Appendix A.

5 Results

5.1 Local leave-one-benchmark-out

Table 2: Brier score by label budget and ALC with one subject per benchmark, mean over the four held-out benchmarks (10 seeds, all rows paired; budgets 3 and 15 in Appendix Table 6). The official baseline is the empirical mean. Each later row adds: shrinkage; a benchmark offset and item pooling (first model); the hedged centre $G/2$ and the false-merge model (hedged-centre model); item groups and the key guard (item-group model); key guard off (final model). *False-merge model*: items sharing a key are pooled, with a probability r that the key wrongly merges items. *Key guard*: sets $r \geq 0.5$ when all of a key's labels are the target subject's. *Step*: ΔALC $\times 10^{-3}$ against the row above, with 95% intervals from a hierarchical bootstrap (benchmarks, then pairs) and a t-interval over the four fold means; the last upper endpoint 0.0 is degenerate (two folds differ by exactly 0). The final-model row replays the shipped archive (platform-like draw: 0.2143). Sources: `src/paiec/analysis/review_round1.py`; `experiments/runs/review_round1/ladder_intervals.json` and `table3_baseline_seeds0-9.json`.

model	B_0	B_1	B_7	B_{31}	ALC	step	hierarchical	t, 4 folds
official baseline	0.250	0.360	0.209	0.189	0.2451			
Beta-binomial shrinkage	0.250	0.231	0.201	0.188	0.2114	-33.70	$[-42.8, -23.4]$	$[-50.6, -16.8]$
first model	0.247	0.229	0.200	0.187	0.2098	-1.62	$[-6.4, +6.4]$	$[-14.1, +10.8]$
hedged-centre model	0.246	0.229	0.200	0.187	0.2097	-0.09	$[-5.2, +3.3]$	$[-8.1, +8.0]$
item-group model	0.246	0.229	0.200	0.184	0.2089	-0.83	$[-1.4, -0.2]$	$[-1.9, +0.2]$
final model	0.246	0.229	0.199	0.183	**0.2088**	-0.09	$[-0.3, 0.0]$	$[-0.3, +0.1]$

Takeaway: shrinkage carries the gain; the refinements are small and not established at benchmark level. *In-sample (four held-out benchmarks).* Shrinkage alone accounts for 33.7 of the total 36.3×10^{-3} gain (Table 2), as empirical Bayes predicts for small per-pair samples [14, 15]. It is the only step established at benchmark level. Everything added afterwards is worth 2.6×10^{-3}, concentrated at budgets 7–31, where item-level evidence accumulates. Its hierarchical interval, $[-4.5, -0.3] \times 10^{-3}$, excludes zero; its t-interval over the four fold means, $[-6.3, +1.1] \times 10^{-3}$, includes zero. On MathArena, Beta-binomial shrinkage beats the final model by 0.67×10^{-3}. Of the single steps, only the item-group step is supported, and only by the hierarchical interval.

Out-of-sample (89 public benchmarks). A pre-registered out-of-sample check uses 89 public benchmarks never used for fitting (2,610 pairs, one subject per benchmark, frozen parameters, no item features). The same benchmarks were used earlier to check the simulator and to reject the τ/mixture variants, so earlier final-model scores on them were known; the pre-registration discloses this. The check confirms the shrinkage step: -32.7×10^{-3} $[-37.1, -28.2]$ (benchmark bootstrap), better on 82 of 89 benchmarks. Its primary outcome, the final model minus shrinkage, is $+0.2 \times 10^{-3}$ $[-0.9, +1.5]$: no difference detected, and inconclusive at the pre-registered $\pm 1 \times 10^{-3}$ minimum effect.

We ship the hierarchical model because it was the pre-declared final model, chosen before the check ran; every paired local step lowers the point estimate; and out of sample no difference from shrinkage was detected. Secondary and descriptive results agree in direction but are not confirmatory. The platform-like (secondary) difference is -1.7×10^{-3} $[-3.1, -0.2]$; of the 89 benchmarks, four derive from training benchmarks, and excluding them the upper end reaches $+0.05 \times 10^{-3}$. Secondary outcomes never override the primary. Of the 89, the 5 with accuracy ≤ 0.15 (a band holding about a third of the platform's pair-slots; Figure 2b) descriptively favour the final model over shrinkage (-7.1×10^{-3}; -13.6×10^{-3} platform-like); there, shrinkage alone is worse than the official baseline by $+9.9 \times 10^{-3}$ $[-1.9, +22.8]$, on 3 of the 5.

5.2 Official platform

Table 3: Local versus platform. (a) Only the first final-model run has per-budget feedback; its SE is *within-draw*, across pairs, not run-to-run. Over four runs the ALC is 0.1839 (SD 0.0240, SE 0.0120, $[0.146, 0.222]$); 1 of 4 runs lies above the local value (sign test $p = 0.625$). (b) Platform: runs and mean with 95% t-interval (fresh draws, unpaired; the item-group archive only, without the re-package in Appendix Figure 3). Replay: in-sample on the replay set, same labels for all versions; exact-set values in parentheses. The replay feeds labels without item features, so it cannot see the item-group step, and its order differs from the local column. On the exact set, baseline minus final is $+0.0165$ $[-0.0032, +0.0352]$ and shrinkage minus final -0.0009 $[-0.0052, +0.0029]$. Sources: `src/paiec/analysis/review_round2.py`; `experiments/runs/review_round2/per_budget_final.json` and `local_vs_platform.json`; `experiments/runs/review_round1/platform_baseline_replay.json`.

(a) final model, Brier by budget

	B_0	B_1	B_3	B_7	B_{15}	B_{31}	ALC
first final-model run	0.2369	0.1903	0.1613	0.1445	0.1463	0.1412	0.1663
within-draw SE (9 pairs)	0.0106	0.0241	0.0268	0.0277	0.0292	0.0290	0.0243
local, one subject per benchmark	0.2456	0.2285	0.2123	0.1995	0.1894	0.1832	0.2088

(b) ALC per version

version	local	platform: n, mean [95% CI]	replay set (exact set)
Beta-binomial shrinkage	0.2114	not submitted	0.1978 (0.1888)
first model	0.2098	1, 0.2157	0.1945 (0.1909)
hedged-centre model	0.2097	5, 0.1923 [0.176, 0.208]	0.1954 (0.1896)
item-group model	0.2089	4, 0.1909 [0.170, 0.212]*	0.1954 (0.1896)
final model	0.2088	4, 0.1839 [0.146, 0.222]	0.1954 (0.1896)

*Includes one run marked Failed but scored (0.1840; ID in Appendix C); without it, $n = 3$, mean 0.1932.

Takeaway: on platform labels, a paired replay confirms the large step (baseline → shrinkage); the later refinements are not distinguishable there, and their order differs from the local ladder. Before its first submission, the re-implemented evaluator predicted 0.212 for our first model, which scored 0.216. Same-archive SDs are 0.010–0.024 across versions (0.013 for the hedged-centre model over 5 runs), of the size small-sample evaluation theory predicts [24, 25] and far larger than the differences between our versions (Table 3; every run in Appendix Figure 3). The platform means do not resolve the order of the versions: intervals overlap, and the order flips depending on whether the run marked Failed is counted. Model choices therefore rest on the paired local tests.

The in-sample replay (§6) scores all versions on the same platform labels of the replay set: the official baseline (0.2276) minus the final model is $+0.0322$ $[+0.0140, +0.0515]$, while Beta-binomial shrinkage minus the final model is $+0.0023$ $[-0.0008, +0.0055]$. There, among the ladder versions, the first model (0.1945) scores best, and on the exact set shrinkage (0.1888) beats the final model (0.1896). Official levels sit below the local ones because platform pairs are more extreme (median accuracy 0.27), which lowers the irreducible $a(1-a)$ (§7). The final ranking uses one common hidden subset, so draw-to-draw variation affects all submissions equally.

5.3 What did not work

Table 4: What we tried and why it was not shipped. Paired $\Delta\mathrm{ALC} \times 10^{-3}$ against the final model with one subject per benchmark, 95% interval over pairs (positive is worse); the adoption gate is ≤ -1.0. "Pre-reg.": yes = pre-registered, post hoc = registered after the run. Appendix Figure 4 plots the first five rows. Sources: `experiments/runs/preregistration/exp_00{42,46,52,53,57}_*.json`; `experiments/registry.csv`.

component	ΔALC [95% CI]	pre-reg.	decision and reason
lexical residual smoothing	$+1.68$ $[+1.08, +2.30]$	yes	dropped: worse
gp-IRT, our reimplementation [3]	$+6.88$ $[+4.80, +8.94]$	yes	not promoted: worse
re-tuned shrinkage strength	$+0.74$ $[-0.28, +1.85]$	yes	no change: mild under-shrinkage only[a]
fitted temperature [20]	$+0.40$ $[+0.24, +0.56]$	yes	stopped: already calibrated[b]
symmetric mixture prior	$+0.52$ $[+0.22, +0.80]$	yes	retired: worse
item scorer	no paired ALC (failed its transfer gate)	yes	dropped
label-free benchmark level	no paired ALC (failed its pre-registered stopping rule)	yes	dropped
nested meta-learned priors	no paired ALC (failed its pre-registered gate)	yes	dropped
prior factorial	no paired ALC (stopped after the pilot)	yes	Laplace-vs-grid audit open[c]
robust, metadata and model-name priors	best gain ≤ 1.0	post hoc	below the gate
smarter label acquisition (4 policies)	best gain ≤ 0.27	post hoc	random acquisition kept
LLM rating distilled into MiniLM/MPNet	no paired ALC (kept $\leq 3\%$ of its ρ_S)	post hoc	dropped

[a] Empirical half-life 5.27 labels against 4.30 for the model. [b] Fitted $T = 0.945$–1.05. [c] Differences 0.012–0.018.

Takeaway: nothing we tried beat the final model. Table 4 lists every candidate with its paired interval, gate and reason. Priors failed for one reason: one to three labels cannot tell prior centres a third of a logit apart, and ability is mostly benchmark-specific. Acquisition failed because the predictor cannot tell unlabelled items apart.

6 Reading per-pair feedback exactly

At budget 0 our model predicts one constant p_0 for every item of a pair. We obtain p_0 by calling each archive's own `predict()` with no labels: 0.448722 for the hedged-centre, item-group and final models, and 0.400099 for the first model. The pair's accuracy a on its scored items then follows from the reported budget-0 Brier score:

$$B_0 = a(1-p_0)^2 + (1-a)p_0^2 \quad \Longrightarrow \quad a = \frac{B_0 - p_0^2}{1-2p_0}. \tag{3}$$

The identity is elementary (the one-bin case of Murphy's decomposition [19]) and has been used independently by others; we contribute error bounds, the reflection pitfall and exact replays (Appendix D). Reported values are rounded to six decimals, with error at most $\eta = 5 \times 10^{-7}$. The reported calibration error (ECE) at budget 0 equals $|a - p_0|$. The two estimates of a therefore agree to within $\varepsilon(p_0) = \eta/|1-2p_0| + \eta$; on every saved pair they agree to within 5.1×10^{-6}. The floor $a(1-a) = B_0 - (p_0 - a)^2$ does not depend on choosing between the two roots. The archive's p_0 is needed only to pick a over $1-a$; getting it wrong flips every accuracy (§7). On *singleton* pairs, whose benchmark appears once in a draw, the prediction stays constant at every budget. Brier score and ECE then fix the number of successes among the revealed labels, so a pair can be replayed exactly in-sample. We use two sets: the *replay set* (18 pairs that replay exactly over our five saved formative runs) and the *exact set* (8 of them, where the item-group and final models also replay exactly; elsewhere item-group effects break the count-only assumption). Appendix A lists every other pair count.

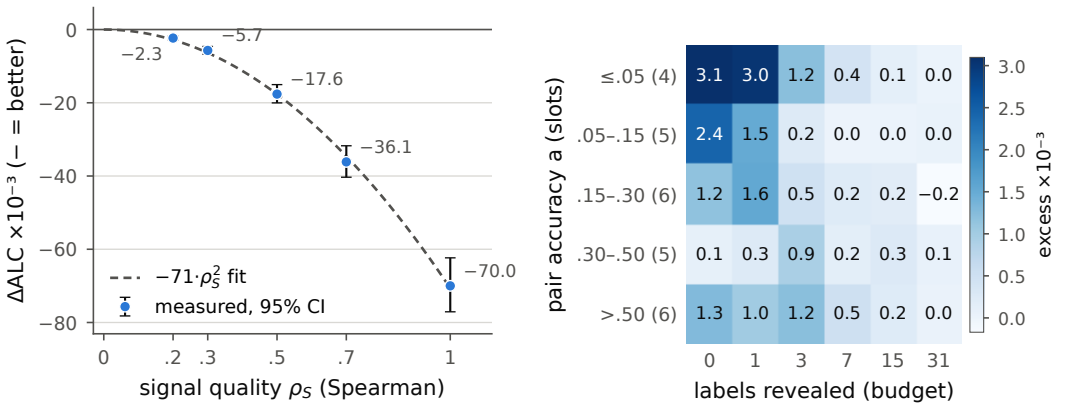
Two replay findings held up under our robustness checks. First, the remaining excess loss is concentrated early and on hard pairs (Figure 2b): 71% of the label-removable excess (the part of the loss above $a(1-a)$ that more labels could remove; Appendix D, Proposition 4) sits at budgets 0–1, and 56% sits on pairs with accuracy ≤ 0.15; whether its cause is the prior centre or the update strength is not settled. Second, no widening of τ or of the prior scale, and not the mixture prior, gives a gain whose interval excludes zero, either in the platform replay or in a replay over 89 public benchmarks: the 92 benchmarks of §7 minus the three original folds among them, on which the shipped parameters were fitted.

7 Simulator validity

A pre-registered check asked whether our local episodes look like the platform's, and it failed. Reading the platform's per-pair feedback through the identity of §6 showed that platform pairs are harder and more extreme than our folds: median pair accuracy 0.27 against 0.41 locally, 14.7% of platform pairs at accuracy ≤ 0.05, and none near 1.

The repair drew from the 92 benchmarks of the enlarged public pool that can produce platform-like episodes, and the check passed: 0 of 11 platform runs outside their 90% interval, a tail count of 6 within the allowed 1–7, and an envelope test $p = 0.544$. The pass survives dropping any single benchmark but not dropping all six largest contributors, so we adopt it only in a narrowed scope: it validates the engine path without item groups, as calibration rather than confirmation, and it predicts a repeat SD of 0.030 against 0.013 observed for the hedged-centre model. An earlier analysis used $p_0 = 0.5513$ instead of 0.448722 and wrongly concluded that the platform was *easier*.

8 Headroom: floor and ceiling bounds



(a) Value of a label-free difficulty signal of quality ρ_S, computed with four named item fields as groups (not the shipped all-field grouping). 95% bands; curve: $-71\rho_S^2$ fit.

(b) Where the hedged-centre, item-group and final models lose on the platform beyond the pair-constant floor $a(1-a)$, by pair-accuracy band and budget (in-sample; slot counts per band in parentheses).

Figure 2: (a) Headroom from item-level signal, measured locally. (b) Platform loss decomposition. Sources: (a) `src/paiec/analysis/e6_headroom.py`; `experiments/runs/e6/e6.json`. (b) `experiments/pattern_mining/meta_2026-10-16/m10_excess_map.py`; `m10_excess_map.csv`.

What item knowledge is worth. Revealing one true quantity at a time to the hedged-centre model shows that item difficulty accounts for about half of the closable budget-0 gap. We had pre-registered an expectation of 60–80% for the benchmark offset. The value of a label-free difficulty signal is roughly quadratic in its quality over $\rho_S \in [0.2, 1.0]$ ($-\Delta\text{ALC}/\rho_S^2 = 58$–$74$; Figure 2a). A scorer at $\rho_S = 0.3$ is worth 5.7×10^{-3}. That is within the range our offline LLM rating reached (0.19–0.34), and above the 0.14–0.23 reported for unseen benchmarks [10]. A runtime LLM item prior, which the rules permit, is therefore the most promising untested direction; we chose a dependency-free, deterministic predictor as a design choice. A scorer must reach $\rho_S \approx 0.5$ to be worth 0.018.

Used without shrinkage the same signal is harmful below $\rho_S \approx 0.5$–0.55, so any external signal needs a learned weight.

Floors. With the true offset, ability and item difficulty all supplied, budget-0 Brier is 0.1016. This is an optimistic bound on one budget, not an ALC floor. A perfect item oracle reaches ALC 0.1387 locally, and nothing we measured shows that the remaining loss is pure noise. On the platform, the *pair-constant floor* $a(1-a)$, the Brier score of knowing each pair's accuracy exactly but nothing about its items, is 0.151 on our saved draws (95% interval clustered by benchmark ID $[0.090, 0.205]$). Locally, the same floor on the four folds is 0.1797 (t-interval $[0.135, 0.224]$), against 0.2088 for the final model.

Ceiling for pair-level methods. The platform floor exceeds the mean ALC of 0.11 that our pre-registration aimed for, but its benchmark-clustered interval includes it; so only on the observed draws does 0.11 require item-level knowledge. On those draws, the largest gain any pair-level adjustment could give is -0.0218 ALC $[-0.0327, -0.0130]$, pooled in-sample over three versions with identical budget-0 predictions. Locally, the gap to the floor is -0.0291.

9 Limitations

- **Evidence base.** Four held-out benchmarks: the item-group step's t-interval over fold means, $[-1.9, +0.2] \times 10^{-3}$ (3 degrees of freedom), includes zero, and with one subject per benchmark cross-subject pooling is inactive. The simulator is valid only for the engine path without item groups, as calibration, and overstates the repeat SD. The oracle of §8 adds independent noise to true difficulty; a real scorer may be worth less.
- **Scope, registration and selection.** The out-of-sample benchmarks have no item features, so the item-group step is unconfirmed there. We did not test runtime LLM priors. 38 of the 60 logged experiments were registered after the fact; the offline LLM rating may have seen the public problems in training. We selected the final model (run 971666).

Acknowledgments and Disclosure of Funding

We thank the organisers for the competition, the data and answering our questions. Code, analyses and this report's text were developed with the help of an LLM-based coding assistant, under the author's direction and review.

References

[1] Stanford AIMS Lab. Predictive AI Evaluation Competition, NeurIPS 2026. https://aims.stanford.edu/competition.

[2] S. Truong and S. Koyejo. AI Measurement Science. Stanford AIMS Lab, 2026.

[3] F. Maia Polo, L. Weber, L. Choshen, Y. Sun, G. Xu, and M. Yurochkin. tinyBenchmarks: evaluating LLMs with fewer examples. In *ICML*, 2024.

[4] A. Kipnis, K. Voudouris, L. M. Schulze Buschoff, and E. Schulz. metabench: a sparse benchmark of reasoning and knowledge in large language models. In *ICLR*, 2025.

[5] R. Vivek, K. Ethayarajh, D. Yang, and D. Kiela. Anchor Points: benchmarking models with much fewer examples. In *EACL*, pages 1576–1601, 2024.

[6] S. Truong, Y. Tu, P. Liang, B. Li, and S. Koyejo. Reliable and efficient amortized model-based evaluation. In *ICML*, 2025.

[7] S. Truong, Y. Tu, R. Schaeffer, and S. Koyejo. Item response scaling laws: a measurement theory approach for efficient and generalizable neural scaling estimation. arXiv:2606.07616, 2026.

[8] V. Hofmann, D. Heineman, I. Magnusson, K. Lo, J. Dodge, et al. Fluid language model benchmarking. In *COLM*, 2025.

[9] E. Shang, W. Tang, and Y. He. LLM evaluation on unseen questions: contextual multidimensional IRT model. arXiv:2608.22295, 2026.

[10] L. Pacchiardi, L. G. Cheke, and J. Hernández-Orallo. 100 instances is all you need: predicting the success of a new LLM on unseen data by testing on a few instances. KDD Workshop on Evaluation and Trustworthiness of Generative AI Models, 2024.

[11] L. Pacchiardi, K. Voudouris, B. Slater, F. Martínez-Plumed, J. Hernández-Orallo, L. Zhou, and W. Schellaert. PredictaBoard: benchmarking LLM score predictability. In *Findings of ACL*, 2025.

[12] J. Park, E. Mendes, G. Stanovsky, and A. Ritter. Anticipatory evaluation of language models. arXiv:2509.20645, 2025.

[13] Y. Ruan, C. J. Maddison, and T. Hashimoto. Observational scaling laws and the predictability of language model performance. In *NeurIPS*, 2024.

[14] B. Efron and C. Morris. Data analysis using Stein's estimator and its generalizations. *Journal of the American Statistical Association*, 70(350):311–319, 1975.

[15] L. D. Brown. In-season prediction of batting averages: a field test of empirical Bayes and Bayes methodologies. *Annals of Applied Statistics*, 2(1):113–152, 2008.

[16] A. Gelman, A. Jakulin, M. G. Pittau, and Y.-S. Su. A weakly informative default prior distribution for logistic and other regression models. *Annals of Applied Statistics*, 2(4):1360–1383, 2008.

[17] G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

[18] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[19] A. H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973.

[20] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *ICML*, PMLR 70:1321–1330, 2017.

[21] J. Kossen, S. Farquhar, Y. Gal, and T. Rainforth. Active testing: sample-efficient model evaluation. In *ICML*, 2021.

[22] T. Zrnic and E. J. Candès. Active statistical inference. In *ICML*, 2024.

[23] W. J. van der Linden and C. A. W. Glas. Capitalization on item calibration error in adaptive testing. *Applied Measurement in Education*, 13(1):35–53, 2000.

[24] S. Bowyer, L. Aitchison, and D. R. Ivanova. Position: don't use the CLT in LLM evals with fewer than a few hundred datapoints. In *ICML*, 2025.

[25] E. Miller. Adding error bars to evals: a statistical approach to language model evaluations. arXiv:2411.00640, 2024.

[26] S. Wang. Measuring all the noises of LLM evals. arXiv:2512.21326, 2025.

A Reproducibility and rerun instructions

Everything runs on CPU. Our environment was Python 3.12.6 on Windows 11 on one laptop (AMD Ryzen 5 5500U, 6 cores, 5.8 GB RAM), with packages pinned in `requirements-lock.txt`. Code release (MIT licence): `https://github.com/Shashwat-srivastav/s2wish-paiec2026` (the release commit is recorded in its history; every released file and its SHA-256 are listed in `RELEASE_MANIFEST.txt`). Seeds are fixed in each module.

Data. The public training data are gated: request access to `aims-foundations/measurement-db` and place the tables under `data/` at the revisions listed in Appendix B. The simulator repair and the 89-benchmark replay also need the 281-benchmark release. Our own formative feedback (`experiments/platform_submissions/`) is not part of the public release. The test of the re-implemented evaluator needs the organisers' official baselines at the revision pinned in `external/PINS.txt`; the engine-equivalence tests need the gated data; the regression tests of the feedback inversion need our feedback files. Tests whose inputs are absent skip rather than fail (`REPRODUCE.md`, §3).

Tests. `python -m pytest -q` runs in about 7 minutes on CPU: 66 tests with the gated data and the organisers' starting kit; 26 in a public clone (18 pass, 8 skip with reasons). The equivalence of the re-implemented evaluator (§3) is tested by `tests/test_simulator.py`, which needs the starting kit. The suite also checks, in `tests/test_v1_3_archive.py`, that the final archive has the quoted checksum, that its code files are byte-identical to the item-group model's, and that its predictions match the research engine's all-field configuration on real episodes (maximum difference below 10^{-9}).

Platform pair sets. Several pair counts appear in the text; each comes from `experiments/runs/review_round1/pair_counts.json` (singleton: the pair's benchmark appears once in its draw; exact: every budget's success count is recovered within $2\varepsilon(p_0)$, uniquely and with nested prefixes).

size	pair set	used in
8	*exact set*: replay-set pairs where the item-group and final models also replay exactly	§5, Table 3
14	exact distinct singleton pairs among the 18 of row (b)	Appendix D; visual atlas
17	the pairs of row (b) within a 0.005 tolerance	visual atlas (replay figure)
18 (a)	*replay set*: exact distinct singleton pairs over our five saved formative runs	§6, §5, Table 3
18 (b)	distinct singleton pairs of our later saved runs (14 exact)	this table
21	exact singleton run-rows over our five saved formative runs	Appendix D
22	distinct singleton pairs over our five saved formative runs	this table (denominator of row 18 (a))
26	pair-slots of the three later models on the earlier draws (floor, ceiling)	§8, Fig. 2b
27	singleton run-rows over our five saved formative runs	Appendix D
34	saved per-pair feedback slots (accuracy recovery check)	§6

item	command	runtime	data
all checks, no data	`python scripts/reproduce_all.py --smoke`	12 s	–
all checks	`python scripts/reproduce_all.py --quick`	82 s	G, F
Table 1	`reproduce_all.py --only T1`	2 s	G
Table 5	`reproduce_all.py --only T2`; rerun `python -m paiec.analysis.item_repr`	<1 s	–; rerun G
Table 2 (R2, seeds 0–9)	`reproduce_all.py --only T3,T3live`	67 s	G
Table 2, official row	`python scripts/table3_official_baseline.py`	30 s	G
Table 2, R1 value	`reproduce_all.py --full --jobs ladder_r1`	not timed	G
Figures 1–4	`python docs/publication/neurips/make_figures.py`	3 s	Fig. 3: F
Table 4, Fig. 4 reruns	`reproduce_all.py --full --jobs exp0042, exp0046,exp0052,exp0053,exp0057`	88 min	G
Figure 2a rerun	`reproduce_all.py --full --jobs e6`	8 min	G
Simulator check	`reproduce_all.py --only SIM`; rerun `python -m paiec.analysis.exp_0039r_simulator_repair`	<1 s	–; rerun G, F
Out-of-sample check (EXP-0058)	`reproduce_all.py --only EXP0058`; rerun `--full --jobs exp0058`	<1 s; rerun 90 s, 242 MB	–; rerun G
Final archive rebuild	`python scripts/build_v1_3_archive.py` (byte-identical)	<1 s	–
Test suite (66 tests)	`python -m pytest -q`	7 min	partly G, F

Archives. SHA-256 checksums:
final model (selected; run 971666), `v1_3_named_facets_noguard.zip`
`a4b0e841d4a344ff207bed2c855342212e95f7d550195b667446ea4e92685b86`

The final archive's `model.py` and `labeling.py` are byte-identical to the item-group model's, and only two parameters differ: `facets` `true`→`"named"` and `guard` $1 \to 0$. The model tests `facets` for truthiness, so `"named"` means *all* fields, despite the archive's name; the effective change is only that the key guard is off. (The research engine's literal `"named"` mode, used for Figure 2a, is a different configuration.) `scripts/build_v1_3_archive.py` rebuilds it byte for byte (with zlib 1.3.x; with another zlib the archive members are still byte-identical). The supplementary research

record gives each registered experiment's question, hypothesis, settings, decision rule, result and commands.

B Training sources

- `aims-foundations/measurement-db`, revision `bc8204d811` (CC BY-SA 4.0, gated): all models and parameters, including the final archive.
- `aims-foundations/measurement-db`, revision `54074453da` (the 281-benchmark release): the expanded-pool prior experiment and the simulator repair. No shipped parameter comes from it.
- `measurement-db-embed` vectors, revision `fb3a155cce` (licence not stated by the provider): the item-representation study (Table 5) only.
- `sentence-transformers/all-MiniLM-L6-v2` (snapshot `1110a243fd`) and `all-mpnet-base-v2` (snapshot `e8c3b32edf`), both Apache 2.0: the distillation experiment only.
- TypeSafe Jev 1.13.0, a hosted service (licence not stated by the provider): offline LLM item ratings and decision checks. It is not used at scoring time.
- Formative platform feedback: used to compare versions and for the diagnostics of §6–§8.

The final archive uses no pretrained model and no network access at scoring time. Table 5 gives the item-representation study cited in §3.

Table 5: ρ_S between predicted and Rasch item difficulty (ridge regression, mean over the five benchmark folds). The LLM rating was computed offline. The rules allow scoring-time calls to approved inference APIs; our shipped predictor makes no network calls by design (no API dependency, deterministic, fully rerunnable). Sources: `src/paiec/analysis/item_repr.py`; `experiments/runs/phase1_item_repr/item_repr_results.csv`.

item representation	ρ_S within a benchmark	ρ_S on an unseen benchmark
length and format	0.22	−0.01
TF-IDF	0.36	−0.08
`measurement-db-embed` vectors	—	−0.08[†]
LLM difficulty rating (offline)	0.22	0.23
all combined	0.41	0.18

[†] Three-benchmark subset; the inner cross-validation chose a constant.

C Names used in this report and internal identifiers

The research record, archive file names and the visual atlas use internal identifiers. This table maps them to the plain names of the main text.

name in this report	internal identifier	archive or record
official baseline (empirical mean)	L0	organisers' starter kit
Beta-binomial shrinkage	v0	`v0.zip`
first model (+ benchmark offset, item pooling)	v1	`v1.zip`
hedged-centre model	v1_1	`v1_1.zip`
item-group model	v1_2	`v1_2.zip`
identical re-package of the item-group model	v1_3 (guard on)	`v1_3_named_facets.zip`
final model	v1_3	`v1_3_named_facets_noguard.zip`
metadata/IRT hybrid	v1_5	`v1_5_hybrid_experimental.zip`
damped early updates	v1_6	`v1_6_damped_facets.zip`
cap on early jumps	v1_7	`v1_7_adaptive_facets.zip`
mixture prior	v1_8, EXP-0057	`v1_8_ext10.zip`
platform-like draw / one subject per benchmark / all subjects	regimes R1 / R2 / R3	
budget-n Brier score	B_n	
item groups / key guard	facets / guard (G1 key guard)	model parameters
false-merge probability r; group variance share λ	ρ; π	model parameters
item scorer; label-free benchmark level; nested meta-learned priors	EXP-0041; EXP-0043; EXP-0045	research record
lexical residual smoothing; gp-IRT reimplementation	EXP-0042; EXP-0046	research record
final-model replay against the research engine	EXP-0044	research record
prior factorial (stopped after pilot)	EXP-0049	research record
re-tuned shrinkage strength; temperature scaling; stress test	EXP-0052; EXP-0053; EXP-0054	research record
simulator validity check; its repair	EXP-0039; EXP-0039R	research record
pre-registered out-of-sample check (§5)	EXP-0058	`experiments/runs/EXP-0058/`
first final-model run; failed-but-scored item-group run	Codabench 967620; 955650	platform feedback
value of an item-difficulty signal (Figure 2a)	E6	`experiments/runs/e6`

Table 6: Table 2 with all six label budgets (Brier score; one subject per benchmark, mean over the four held-out benchmarks, 10 seeds, paired). Sources: `experiments/runs/review_round1/ladder_intervals.json` and `table3_baseline_seeds0-9.json`.

model	B_0	B_1	B_3	B_7	B_{15}	B_{31}	ALC
official baseline	0.250	0.360	0.243	0.209	0.194	0.189	0.2451
Beta-binomial shrinkage	0.250	0.231	0.214	0.201	0.192	0.188	0.2114
first model	0.247	0.229	0.213	0.200	0.191	0.187	0.2098
hedged-centre model	0.246	0.229	0.213	0.200	0.191	0.187	0.2097
item-group model	0.246	0.229	0.212	0.200	0.190	0.184	0.2089
final model	0.246	0.229	0.212	0.199	0.189	0.183	0.2088

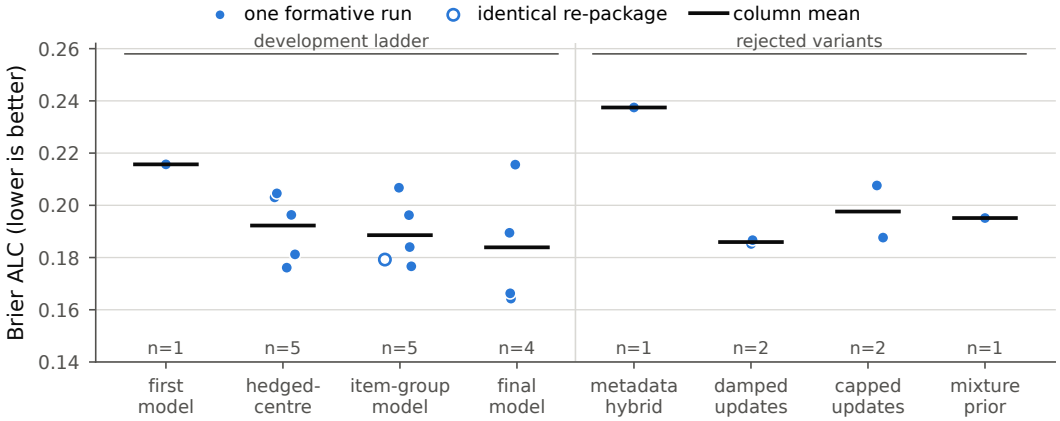


Figure 3: Logged formative runs of the development ladder and the rejected variants (one point each, fresh draws; bars are column means). *Development ladder*: the models of Table 2; the open circle is a functionally identical re-package, so the item-group column has $n = 5$ (Table 3 counts the item-group archive only, $n = 4$). *Rejected variants*: a metadata/IRT hybrid, damped early updates, a cap on early jumps, the mixture prior. Runs are unpaired: the final model alone ranges from 0.164 to 0.216. Sources: `docs/publication/neurips/make_figures.py`; `experiments/platform_submissions/*.json`.

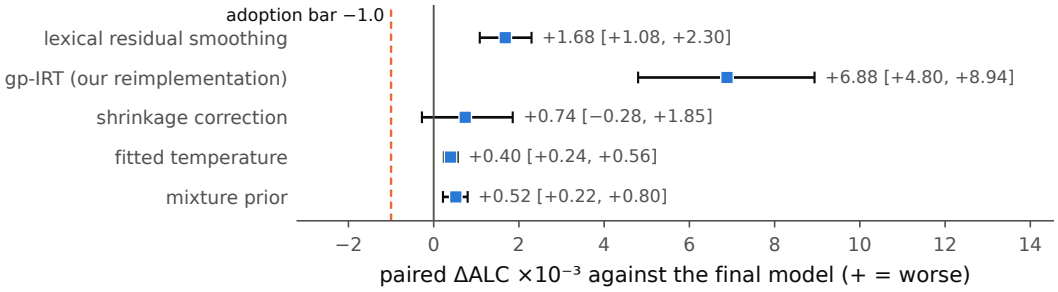


Figure 4: The five pre-registered candidates of Table 4, each paired against the final model with one subject per benchmark ($\Delta\text{ALC} \times 10^{-3}$, 95% interval over pairs; positive is worse). The dashed line is the pre-registered adoption bar (-1.0). None qualifies. Sources: `make_figures.py`; `experiments/runs/preregistration/exp_00{42,46,52,53,57}_*.json`.

D Exact inversion of per-pair formative feedback

The platform's formative feedback reports, for every evaluated subject–benchmark pair of our own submission, the Brier score B_b and the calibration error E_b at each label budget $b \in \mathcal{B} = \{0, 1, 3, 7, 15, 31\}$, rounded to six decimals. This appendix shows that, under conditions we state and check, these numbers determine (a) the pair's accuracy, (b) on singleton pairs, the number of successes among the revealed labels, and hence (c) the exact score of any pair-level change to the model, together with floor and ceiling bounds. The implementation is the NumPy package `paiec.feedback`; each number quoted here is asserted by `tests/test_feedback_inversion.py`.

Setting. A pair has a fixed evaluation pool of scored response rows with outcomes $y_i \in \{0, 1\}$, the same at every budget; its *accuracy* is $a = \bar{y}$, the mean outcome over the scored rows. Labels are revealed from a separate acquisition pool; at budget b the model has seen the first b of them, of which k_b are successes. The scoring rules are those published by the organisers: $B_b = \frac{1}{n}\sum_i (p_i - y_i)^2$, and $E_b = \sum_j \frac{n_j}{n} |\bar{p}_j - \bar{y}_j|$ over ten fixed equal-width bins $[0, 0.1), \ldots, [0.9, 1]$ weighted by response count; the pair's ALC is $\sum_b w_b B_b$ with $w = (0.1, 0.2, 0.2, 0.2, 0.2, 0.1)$. We assume:

- A1. *Constant budget-0 prediction:* the model predicts one value p_0 on every scored row of the pair. We obtain p_0 by calling the archive's own `predict()` with no labels (and check it is constant across items and subjects); it is never read off the feedback.
- A2. *Exact scoring:* the reported values are the rules above applied to binary outcomes.
- A3. *Reported precision:* each reported value has absolute rounding error at most $\eta = 5 \times 10^{-7}$.
- A4. *Count-only prediction (for k only):* at budget b the prediction on every scored row is $p(k_b, b)$, a function of the success count alone. For our hierarchical models this holds on *singleton pairs* (the benchmark appears once in the draw, so no item pooling across subjects) whose items trigger no item-level term.
- A5. *Nested prefixes (for k only):* with accept-all acquisition the platform reveals labels in a fixed order, so $0 \le k_{b'} - k_b \le b' - b$ for $b < b'$, and a pair has the same k path in every run.

Lemma 1 (constant-prediction identities). If a pair's prediction is a constant q, then $B = a(1 - 2q) + q^2 = a(1 - a) + (q - a)^2$ and $E = |q - a|$, whatever the bin edges. *Proof.* Since $y_i^2 = y_i$, $\frac{1}{n}\sum_i (q - y_i)^2 = q^2 - 2q\bar{y} + \bar{y}$, which rearranges to both forms. All rows fall in one bin, whose gap is $|q - \bar{y}|$ with weight 1. □

Proposition 1 (accuracy). Under A1–A2 and $p_0 \neq \frac{1}{2}$,

$$a = \frac{B_0 - p_0^2}{1 - 2p_0}, \qquad \text{and independently} \qquad a \in \{p_0 - E_0,\ p_0 + E_0\}. \tag{4}$$

Under A3 the first expression has error at most $\eta/|1 - 2p_0|$, so it and the nearer ECE root disagree by at most $\varepsilon(p_0) = \eta/|1 - 2p_0| + \eta$. *Proof.* Solve Lemma 1 at $b = 0$ for a; the error bound is the derivative $1/(1 - 2p_0)$ times η, plus η for E_0. □

The bound shows where the method is fragile: the condition number $1/|1-2p_0|$ is about 10 at our $p_0 = 0.448722$, so $\varepsilon = 5.38 \times 10^{-6}$ there (3.00×10^{-6} at the first model's $p_0 = 0.400099$), and it diverges at $p_0 = \frac{1}{2}$.

Proposition 2 (identifiability and the reflection pitfall). Without p_0, the pair (B_0, E_0) identifies only $a(1-a) = B_0 - E_0^2$, i.e. the unordered set $\{a, 1-a\}$; the points $(a, a \pm E_0)$ and $(1-a, 1-a \pm E_0)$ all reproduce (B_0, E_0). In particular the reflection $\rho : (a, p_0) \mapsto (1-a, 1-p_0)$ leaves both reported values unchanged, so substituting $1-p_0$ for p_0 in (4) returns $1-a$ *and still passes the ECE cross-check*. At $p_0 = \frac{1}{2}$, $B_0 = \frac{1}{4}$ for every a and only $|a - \frac{1}{2}|$ is identified. *Proof.* By Lemma 1, $B_0 - E_0^2 = a(1-a)$, which fixes a up to $a \leftrightarrow 1-a$, and $p_0 = a \pm E_0$. Under ρ, $(1-a)a + ((1-p_0) - (1-a))^2 = a(1-a) + (a-p_0)^2$ and $|(1-p_0) - (1-a)| = |a - p_0|$. □

A cautionary example (our own error). Until 2026-10-15 we used $p_0 = 0.5513$, which is $1 - 0.448722$ to four decimals, instead of calling the archives. Every recovered accuracy was reflected and the ECE check passed on every pair. The reflected reading put the median pair accuracy of our 34 saved pair-slots at 0.73, with none at or below 0.05. The correct reading gives 0.27, with 14.7% at or below 0.05: platform pairs are harder than our local folds, not easier. Quantities symmetric in a, such as Brier scores, ALC and the floor below, were unaffected. The lesson is general: the orientation of the inversion must come from the model, never from the feedback.

Proposition 3 (success counts on singleton pairs). Under A1–A5, $E_b = |p(k_b, b) - a|$, so k_b minimises the residual $r_b(k) = \big||p(k,b) - a| - E_b\big|$ over $k = 0, \ldots, b$, where $p(k,b)$ is computed by calling the model with k successes among b synthetic labels. We accept a recovery as *exact* if $\min_k r_b(k) \le 2\varepsilon(p_0)$ and *unique* if every other k has residual above $2\varepsilon(p_0)$. Recovery is not unique when two counts give predictions at the same distance on either side of a, or predictions closer together than the tolerance. A5 provides two independent checks: the path must be nested, and it must be identical across runs. Lemma 1 provides a third: $B_b - a(1-a) - E_b^2 = 0$ whenever the prediction is constant. *Proof.* Lemma 1 at budget b with $q = p(k_b, b)$, and the error bound of Proposition 1 applied to a and E_b. □

Corollary (exact zero-upload replay). Let $q(k,b)$ be any *pair-level* rule, i.e. one value per pair from the revealed count. Given a and the recovered path (k_b), its score on the platform pair is exactly $B_b(q) = a(1-a) + (q(k_b, b) - a)^2$, with ALC $\sum_b w_b B_b(q)$. No upload, rerun or fitting is involved. At budget 0 only a is needed, so budget-0 counterfactuals are exact on every pair where the observed and the alternative budget-0 predictions are constant.

Proposition 4 (floor and ceiling). For any pair-level method, $B_b \ge a(1-a)$, with equality if and only if $q = a$. Hence the mean ALC over pairs is at least the *pair-constant floor* $F = \overline{a(1-a)}$, which the feedback identifies even without p_0 (as $B_0 - E_0^2$). With the *label-removable excess* $X_b = \overline{B_b - a(1-a)}$ of the observed curves, no pair-level change can improve ALC by more than $\sum_b w_b X_b = \overline{\mathrm{ALC}} - F$. *Proof.* Lemma 1 and $\sum_b w_b = 1$.□ Methods whose predictions vary across items within a pair, such as item pooling across subjects, are not covered and can go below F.

Validation. *Synthetic.* We simulated pairs, scored them with the published rules (which match our re-implemented evaluator to 10^{-12}), rounded to six decimals and inverted them. Over 5,000 draws, excluding those with $|1-2p_0| < 0.02$, the error of (4) never exceeded $\eta/|1-2p_0|$. With beta-binomial pair-level rules we generated 1,310 pairs (6,550 budget steps). The true k_b was always within tolerance, and 6,549 steps had a unique minimiser, which was always correct (the one non-unique step was a tie between two counts). Both self-replay and replay of an alternative rule matched direct simulation within 2ε. Item-varying predictions were flagged by the constancy residual. *Our saved feedback* (Table 7). The 34 saved pair-slots come from four runs of four archives. The exact k paths of 18 distinct singleton pairs (21 run-rows of 27 singleton rows; 14 pairs without the first model's run) satisfied every check: residual below 5×10^{-6}, nested, identical in the 3 pairs seen in two runs, and a self-replay that reproduces every reported B_b within 3.2×10^{-6}. The remaining singleton rows had residuals of at least 5.1×10^{-4}, two orders of magnitude above the largest exact residual, which is the expected footprint of item-level terms violating A4. We therefore do not use them.

Table 7: Validation of the inversion on our own saved feedback. Bounds are from Proposition 1 with $\eta = 5 \times 10^{-7}$.

quantity	value	check
max $\lvert a - a_{\mathrm{ECE}} \rvert$, 34 saved pair-slots	5.05×10^{-6}	$\leq \varepsilon = 5.38 \times 10^{-6}$
B_0 reconstructed from (a, p_0), 34 slots	error 2.8×10^{-17}	Lemma 1
pair-constant floor F, 26 slots of the three later models (v1_1–v1_3)	0.1511	Prop. 4
label-removable excess $X_0, \ldots, X_{31}$ (same 26)	0.081, 0.037, 0.021, 0.007, 0.004, 0.000	Prop. 4
ceiling for any pair-level change (same 26)	-0.0218 ALC	observed ALC 0.173

Limitations. (i) Every replay is in-sample. It scores a change on the draw that informed it, and platform draws differ (the floor is 0.151 on the earlier draws). Replays describe a draw; they do not predict the summative score. (ii) Recovering k is exact only on singleton pairs with count-only predictions; on our runs this leaves 18 pairs, so pair-level conclusions rest on small samples. (iii) Exactness is exactness up to the reported precision. The bounds assume six-decimal rounding and degrade as $1/|1 - 2p_0|$, and a platform that rounds more coarsely, or a model with p_0 near $\frac{1}{2}$, would loosen or void them. (iv) The method needs the submitted model to be callable offline, to obtain p_0 and $p(k, b)$. It does not apply to item-varying models without further assumptions.

Ethics and competition rules. The method reads only the formative feedback that the platform shows a team about its own submissions, together with that team's own archives. It never accesses any feedback other than our own, nor the hidden data, and it never tries to determine which real subject or benchmark lies behind an anonymous ID. Throughout, the IDs serve only as opaque grouping labels, and we name none. Every use of feedback in this report is diagnostic.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the main finding and its scope: shrinkage carries most of the gain, the refinements are not distinguishable from zero across the four held-out benchmarks, and the remaining gap is item-level (§5, §8).

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 9 (Limitations) covers simulator validity, the four folds, platform noise, oracle assumptions, the scope of the out-of-sample check and post-hoc registration; the unreached 0.11 aspiration is discussed in §8.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The only formal results are the feedback-inversion identity and its corollaries; their assumptions (A1–A5) and proofs are in Appendix D.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Appendix A gives the environment, commands, seeds, data revisions and the checksum of every submitted archive, including the selected submission (the final model, Codabench run 971666); Appendix B lists every training source; §4 gives all fitted parameters. The out-of-sample check EXP-0058 (§5) was pre-registered with its benchmark list, arms, seeds and decision rule (`docs/preregistrations/EXP-0058_oos_confirmation_preregistration.md`) and runs in about 90 s.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.

(c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The code is public at https://github.com/Shashwat-srivastav/s2wish-paiec2026. The training data are gated by the organisers; Appendix A says how to request access and where to place them.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Folds, episode regimes and seeds are in §3; priors, fitted values and integration in §4; hyperparameters were fitted on the public benchmarks only (Appendix B).

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Comparisons are paired on identical episodes with 95% bootstrap intervals over pairs; benchmark-level claims use t-intervals over the four fold means; platform runs are reported individually with their SDs (§3, §5). The pre-registered out-of-sample check EXP-0058 reports percentile benchmark-level bootstrap intervals over its 89 held-out benchmarks (10,000 resamples), with its primary and secondary outcomes fixed in advance (§5). Bootstraps are percentile intervals; no normality is assumed except for the t-intervals over fold means and platform runs.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Every experiment ran on one laptop CPU, an AMD Ryzen 5 5500U (6 cores, 12 threads) with 5.8 GB of RAM visible to Windows 11, and no GPU (Appendix A). Measured wall times (`REPRODUCE.md`): the test suite takes 419 s; checking every reported number from saved results takes 82 s; rerunning the heavy experiments one at a time takes about 2.5 h; the five experiments of Figure 4 took 1971, 353, 727, 1795 and 463 s. The predictor takes 3.3–222 ms per call (§4). The total compute of the whole programme, including failed experiments, was not logged; from these runtimes we estimate it at tens of CPU-hours (an estimate, not a measurement).

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The work uses public, organiser-provided benchmark results; no human subjects or personal data are involved, and anonymous benchmark IDs were not de-anonymised (Appendix D).

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [No]

Justification: The paper has no separate impact section. Cheaper evaluation is the intended benefit; the main risk is using predicted scores in place of real evaluation, which could hide failures on a new benchmark. Our intervals and Limitations section quantify how uncertain such predictions are.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: We release a 5 KB statistical predictor and analysis code; neither poses a high risk of misuse.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: Every asset is credited with its revision and licence in Appendix B. Licences as stated on the providers' Hugging Face cards (checked 2026-10-22): `aims-foundations/measurement-db`, CC BY-SA 4.0 (gated; used under its access terms and not redistributed); `sentence-transformers/all-MiniLM-L6-v2` and `all-mpnet-base-v2`, Apache 2.0. For `aims-foundations/measurement-db-embed` and the hosted TypeSafe Jev service, a licence is not stated by the provider.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The released code is documented in its README and release manifest, and the feedback-inversion package has its own documentation and tests. Licence: the code release is under the MIT License (`LICENSE` in the repository).

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: No crowdsourcing or human-subject research was involved.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human subjects were involved.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: An LLM (TypeSafe Jev 1.13.0) produced offline item-difficulty ratings and decision checks used in our experiments (Table 5, §8, Appendix B); it is not used at scoring time, by design (the rules permit approved inference APIs at scoring time; our predictor makes no network calls). An LLM-based coding assistant helped develop code and text (Acknowledgements).

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.